Test 2 is Friday. Today: every question type, once.

- **Turn in Quiz 9** now.
- Test 2 covers **4.2** (exponentials and interest), **4.3** (logarithms) and **geometry** (polygons and circles).
- For each type: I work a model, then you try one. Answers appear when we check.
- Bring a scientific calculator on Friday, and show your work for full credit.

Printed on Test 2

- $A = P(1 + r/n)^{nt}$
- $A = Pe^{rt}$
- Heron's formula

From memory

- log ↔ exponential form, change of base
- DMS, degrees ↔ radians
- arc length, sector area, circle angle rules
- triangle and trapezoid area, proportional segments
- $v = r\omega$

SECTION 4.2 · INTEREST TABLE

Compounded n times per year, and continuously

$A = P(1 + r/n)^{nt}$ · $A = Pe^{rt}$ · r as a decimal · round to the nearest cent

Model: $3,000 at 4% for 15 years

n	12	365	continuous
A	$5,460.90	$5,466.18	$5,466.36

e.g. $3000(1 + 0.04/12)^{180} \approx 5460.90$

You try: $1,800 at 5.5% for 8 years

n	12	365	continuous
A	$2,792.06	$2,794.78	$2,794.87

The log is the exponent

$$\log_b x = y \iff b^y = x$$

Model

$16^{3/4} = 8 \rightarrow \log_{16} 8 = 3/4$

$\log_9 243 = 2.5 \rightarrow 9^{2.5} = 243$

You try

$27^{2/3} = 9 \rightarrow \log_{27} 9 = 2/3$

$\log_4 1024 = 5 \rightarrow 4^5 = 1024$

Undo the log, or take the log

$\ln x = c \to x = e^c \cdot b^x = c \to x = \ln c \div \ln b$

Model

$\ln x = 2.7 \to x = e^{2.7} \approx 14.880$

$3^x = 20 \to x = \ln 20 \div \ln 3 \approx 2.727$

You try

$\ln x = 3.2 \to x = e^{3.2} \approx 24.533$

$5^x = 17 \to x = \ln 17 \div \ln 5 \approx 1.760$

DMS both directions

DMS → decimal: D + M/60 + S/3600 (four places) · decimal → DMS: × 60, then × 60 again (nearest second)

Model

$52°17′48″ \to 52 + 17/60 + 48/3600 \approx 52.2967°$

$41.6375° \to 0.6375 \times 60 = 38.25′$; $0.25 \times 60 = 15″ \to 41°38′15″$

You try

$18°05′33″ \to 18.0925°$

$73.2186° \to 73°13′7″$

Multiply by π/180 or 180/π

degrees → radians: × π/180 · radians → degrees: × 180/π

Model

$1.9 \text{ rad} \to 1.9 \times 180/\pi \approx 108.9°$

$147.3° \to 147.3 \times \pi/180 \approx 2.57 \text{ rad}$

You try

$4.1 \text{ rad} \to \approx 234.9°$

$58.6° \to \approx 1.02 \text{ rad}$

Fraction of the whole circle

$$s = (\theta/360) \cdot 2\pi r \cdot A = (\theta/360) \cdot \pi r^2$$

Model: r = 6.4, θ = 110°

$$s = (110/360)(2\pi)(6.4) \approx 12.3$$

$$A = (110/360)\pi(6.4)^2 \approx 39.3$$

You try: r = 8.2, θ = 50°

$$s \approx 7.2$$

$$A \approx 29.3$$

Use θ directly

$s = r\theta \cdot A = \frac{1}{2} r^2 \theta$

Model: r = 7, θ = 2π/5

$$s = 7(2\pi/5) = 2.8\pi \approx 8.8$$

$$A = \tfrac{1}{2}(49)(2\pi/5) = 9.8\pi \approx 30.8$$

You try: r = 12, θ = 5π/6

$$s = 10\pi \approx 31.4$$

$$A = 60\pi \approx 188.5$$

Three sides, or a right triangle

Heron (printed): $s = (a + b + c)/2$, $A = \sqrt{(s(s - a)(s - b)(s - c))}$ · right triangle: $A = ½ \cdot \text{leg} \cdot \text{leg}$

Model

$a = 1.8$, $b = 2.4$, $c = 3.1$ cm
$s = 3.65$; $A = \sqrt{(3.65 \cdot 1.85 \cdot 1.25 \cdot 0.55)} \approx 2.2$ cm^2

Right triangle: hypotenuse 6.5 cm, leg 2.5 cm
other leg $= \sqrt{(6.5^2 - 2.5^2)} = 6.0$; $A = ½(2.5)(6.0) = 7.5$ cm^2

You try

$a = 2.6$, $b = 1.9$, $c = 3.3$ cm
$s = 3.9$; $A \approx 2.5$ cm^2

Right triangle: hypotenuse 5.3 cm, leg 2.8 cm
other leg $= 4.5$; $A = 6.3$ cm^2

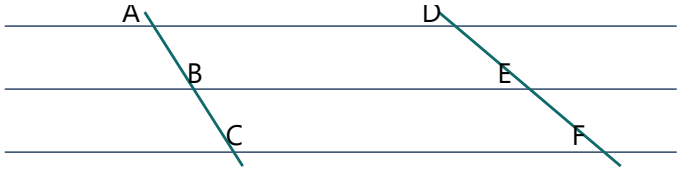
Parallel sides and parallel lines

Trapezoid: $A = \frac{1}{2} h (b_1 + b_2)$

Model: bases 3.4 and 7.8 cm, height 2.6 cm → $\frac{1}{2}(2.6)(11.2) = 14.56$ cm²

You try: bases 4.2 and 9.5 cm, height 3.1 cm → 21.24 cm²

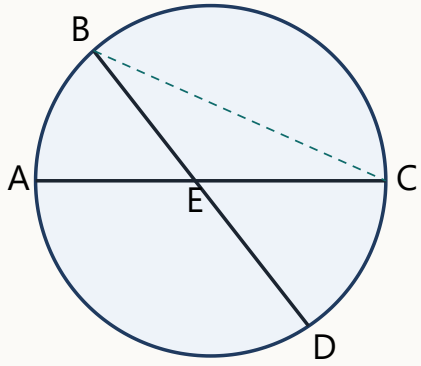
Three parallel lines cut two transversals: AB/BC = DE/EF



Model: AB = 4.5, BC = 6.0, DE = 3.6 → EF = $3.6(6.0)/4.5 = 4.8$

You try: AB = 5.2, BC = 3.9, EF = 2.7 → DE = $2.7(5.2)/3.9 = 3.6$

Angles and arcs inside a circle



AC is a diameter. Chord BD crosses it at E.
Rules: ∠AEB + ∠AED = 180° · inscribed angle = ½ arc · ∠AEB = ½(arc AB + arc CD)

Model: ∠AED = 128°, arc AB = 48°

$$\angle AEB = 180 - 128 = 52^\circ \cdot \angle ACB = \tfrac{1}{2}(48) = 24^\circ \cdot \text{arc CD} = 2(52) - 48 = 56^\circ$$

You try: ∠AED = 140°, arc AB = 30°

$$\angle AEB = 40^\circ \cdot \angle ACB = 15^\circ \cdot \text{arc CD} = 50^\circ$$

The unit chain, one more time

The blade length is the radius. v = (rpm)(2π)(r) gives feet per minute; × 60 ÷ 5280 gives miles per hour.

Model: blade 2.5 ft, 400 rpm

$$v = 400(2\pi)(2.5) = 2000\pi \approx 6283.2 \text{ ft/min}$$

$$6283.2 \times 60 \div 5280 \approx 71.4 \text{ mi/h}$$

You try: blade 3.5 ft, 300 rpm

$$2100\pi \approx 6597.3 \text{ ft/min}$$

$$\approx 75.0 \text{ mi/h}$$

Where points are usually lost

- Using the rate as a percent (4) instead of a decimal (0.04).
- Moving the base to the wrong side when switching log ↔ exponential form.
- Rounding in the middle of a calculation.
- Using degrees in $s = r\theta$ or $A = \frac{1}{2}r^2\theta$.
- Forgetting the seconds step in decimal → DMS.
- Using the diameter (or the whole trapezoid side) where the formula wants a radius or height.
- Matching the wrong segments in a proportion.
- Leaving off units: cm^2, ft/min, mi/h.

Before Friday

Tonight and Thursday

- Work the Test 2 practice PDF on the course page without notes, then check it.
- Fill in the formula page from memory.
- Take the Test 2 readiness check on the review page until you reach 80%.
- Redo any problem type from today that you missed.

Friday, October 30

- Test 2, in class on paper
- Scientific calculator
- Show your work for full credit
- Compound interest, continuous interest and Heron's formula are printed; everything else is from memory